Intermediate Microeconomics

Week 2 · Day 3 — Competitive Firms and Markets (Perloff, Chs. 8, 8–9)

The price-taking firm, its supply curve, and long-run entry

Recap of Day 2, and today's plan

Day 2 measured welfare in our market $Q_d = 120 - 2P$, $Q_s = -60 + 4P$:

$$P^* = 30, \quad Q^* = 60, \qquad CS = 900, \quad PS = 450, \quad TS = 1350.$$

We treated supply as given. **Today we open it up**: that supply curve is 20 identical firms, each choosing output to maximise profit. We will rebuild $Q_s = -60 + 4P$ from the firm up, then let firms enter.

Plan

The price-taking firm	why it takes price as given; its profit problem
Cost curves & the shutdown rule	MC, AVC, AC; operate or close in the short run
The firm's supply curve	from $P = MC$, aggregated to the market
Entry, exit, long run	profit $\Rightarrow$ entry $\Rightarrow$ zero-profit equilibrium

Learning goals — Day 3

1. State the assumptions of perfect competition and why they make the firm a price taker.
2. Set up and solve the competitive firm's profit-maximisation problem ($P = MC$).
3. Build MC, AVC and AC from a cost function and locate the shutdown and break-even prices.
4. Derive the firm's short-run supply curve and aggregate it to the market.
5. Link producer surplus, profit, and fixed cost at the firm level.
6. Trace the long run: positive profit $\Rightarrow$ entry $\Rightarrow$ the zero-profit number of firms.

Firms in a competitive market: price takers

A competitive firm is small relative to its market: it can sell as much as it likes at the going price P, and *nothing* at a price above it (buyers switch to identical rivals).

The firm's demand curve is flat

Each firm faces a **horizontal** demand curve at the market price — perfectly elastic. It chooses *quantity*, not price. Contrast the *market* demand, which slopes down.

This is an idealisation, but a useful one: it is the benchmark of an efficient market (Day 2) and the foil for monopoly (Day 4). The wheat grower, the corner currency kiosk, and the single Uber driver are close enough to price takers for the model to bite.

Perfect competition: the four assumptions

Properties

1. **Many buyers and sellers**, each tiny — no one moves the price.
2. **Homogeneous product** — buyers see rivals' goods as identical, so no price premium survives.
3. **Free entry and exit** — no barriers in the long run (the engine of today's second half).
4. **Perfect information** and negligible transaction costs.

Assumptions 1–2 give price taking (short run). Assumption 3 pins down the long run. Drop assumption 1 and you get monopoly (Day 4); drop 2 and you get monopolistic competition.

Profit maximisation: total and marginal revenue

The firm chooses q to maximise profit

$$\pi(q) = \underbrace{P\,q}_{TR} - C(q).$$

Marginal revenue equals price

Because P is fixed to the firm, $TR = Pq$ is a straight line through the origin, so

$$MR = \frac{\mathrm{d}TR}{\mathrm{d}q} = P.$$

This is the special feature of competition: $MR = P$. (For the monopolist on Day 4, $MR < P$.)

Selling one more unit adds exactly P to revenue — no need to cut price on the other units, because the firm is too small to move P.

The optimality condition: $P = MC$

Maximise $\pi(q) = Pq - C(q)$. First-order condition:

$$\pi'(q) = P - MC(q) = 0 \implies \boxed{P = MC(q)}.$$

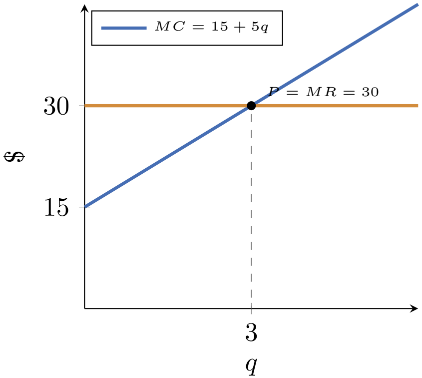
Second-order condition $\pi''(q) = -MC'(q) < 0$ needs *MC rising*.

Reading the rule

Produce until the cost of the last unit just equals the price it fetches. Below that q, $P > MC$ (expand); beyond it, $P < MC$ (contract).

For our firm $C(q) = 22.5 + 15q + 2.5q^2$, $MC = 15 + 5q$. At $P = 30$:

$$30 = 15 + 5q \Rightarrow q = 3.$$



The firm's cost curves

From $C(q) = 22.5 + 15q + 2.5q^2$:

$$MC = 15 + 5q, \quad AVC = \frac{VC}{q} = 15 + 2.5q,$$

$$AC = \frac{22.5}{q} + 15 + 2.5q.$$

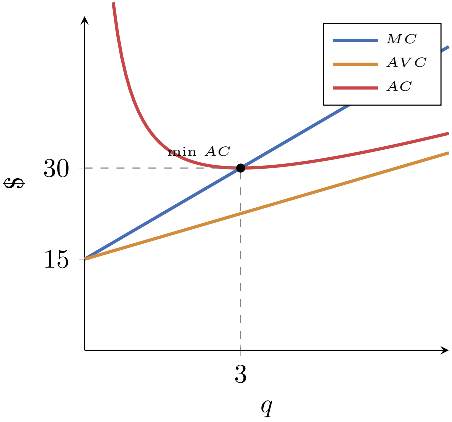
Minimum AVC

AVC falls to 15 as $q \to 0$: the **shutdown price** is 15.

Minimum AC

$AC'(q) = -22.5/q^2 + 2.5 = 0 \Rightarrow q = 3$, giving $AC(3) = 30$: the **break-even price** is 30.

Numeric check: At $q = 3$, $MC = AC = 30$ — MC cuts AC at its minimum.



Operate or shut down? The short-run rule

In the short run fixed costs are sunk. The firm compares the price to *average variable* cost:

Shutdown rule

$$\text{Operate if } P \geq \min AVC = 15; \qquad \text{shut down } (q = 0) \text{ if } P < 15.$$

Three regimes for our firm:

- $P \geq 30$: price covers *average* cost — **profit** ≥ 0.
- $15 \leq P < 30$: price covers variable but not full cost — **operate at a loss** (losing less than the sunk fixed cost of 22.5).
- $P < 15$: revenue cannot even cover variable cost — **shut down**.

Producing at a loss is rational: at $P = 25$, $q = 2$, revenue 50 exceeds variable cost 40, contributing 10 toward the fixed cost the firm owes either way.

Deriving the firm's short-run supply curve

Exercise. Find the firm's short-run supply, then aggregate over the 20 firms.

Step 1 — apply $P = MC$

$$P = 15 + 5q.$$

Step 2 — invert for q

$$q = \frac{P - 15}{5} \qquad (\text{for } P \geq 15).$$

Step 3 — impose the shutdown rule

$$q_s(P) = \begin{cases} \dfrac{P-15}{5}, & P \geq 15 \\ 0, & P < 15. \end{cases}$$

Step 4 — aggregate horizontally over 20 firms

$$Q_s = 20 \cdot \frac{P - 15}{5} = 4(P - 15) = 4P - 60 = -60 + 4P.$$

Numeric check: This is exactly Day 1's market supply. The supply curve was always 20 firms' MC curves, summed.

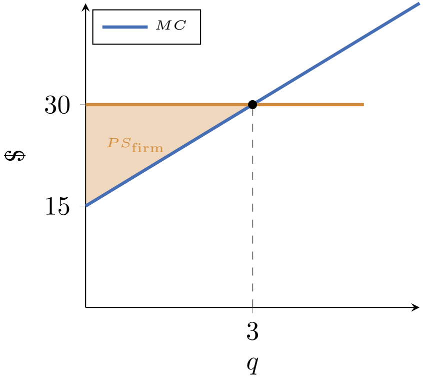
Producer surplus, profit, and fixed cost

At $P = 30$ each firm produces $q = 3$. Its **producer surplus** is the area between price and MC:

$$PS_{\text{firm}} = \tfrac{1}{2}(30 - 15)(3) = 22.5.$$

Its **profit** nets out fixed cost:

$$\pi = PS_{\text{firm}} - FC = 22.5 - 22.5 = 0.$$



Two facts to keep

- $PS = \pi + FC$: surplus exceeds profit by the fixed cost.
- Market $PS = 20 \times 22.5 = 450 =$ the Day 2 figure. The big triangle *is* the sum of firm rectangles-minus-MC.

Positive profit is possible — a demand boom

Start in long-run equilibrium: 20 firms, $P = 30$, each earning *zero* profit. Now incomes rise and demand shifts out to $Q'_d = 150 - 2P$ (Day 1's shock).

Step 1 — short run: $n = 20$ is fixed

Market supply is still $Q_s = 4P - 60$:

$$150 - 2P = -60 + 4P \Rightarrow P = 35,\ Q = 80.$$

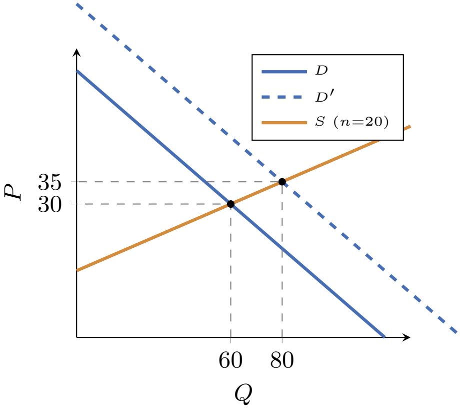
Step 2 — each firm expands

$$q = \frac{35 - 15}{5} = 4.$$

Step 3 — profit turns positive

$$\pi = 35(4) - \big[22.5 + 60 + 40\big] = 140 - 122.5 = 17.5 > 0.$$

Numeric check: Price $35 > \min AC = 30$, *so every firm is now above break-even.*



Entry competes the profit away: how many firms?

Positive profit is a signal: with free entry, new firms come in, shifting market supply *right* and pushing price *down*. Entry stops only when profit is back to zero — i.e. when price returns to min $AC = 30$.

Step 1 — long-run price is pinned

Entry drives $P \to \min AC = 30$; each firm returns to $q = 3$.

Step 2 — meet the boosted demand at $P = 30$

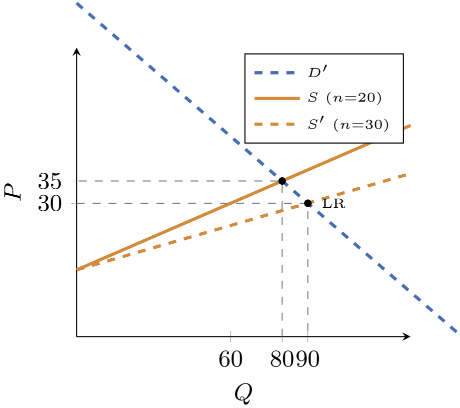
$$Q_d' = 150 - 2(30) = 90.$$

Step 3 — count the firms

$$n = \frac{Q}{q} = \frac{90}{3} = 30.$$

So 10 new firms enter ($20 \to 30$), each again earning *zero* profit.

Numeric check: Check: 30 *firms* $\times\, q = 3 = 90 = Q_d'(30)$. ✓



The long-run rule, and the flat long-run supply

A firm's long-run decision

$$\text{Enter / stay if } P \geq \min AC = 30; \qquad \text{exit if } P < 30.$$

In the long run all costs are avoidable, so the relevant threshold rises from $\min AVC = 15$ (short run) to $\min AC = 30$ (long run).

Because entry and exit peg the price at $\min AC = 30$ regardless of demand, the **long-run supply curve is horizontal at** $P = 30$ (a constant-cost industry). Demand booms raise the *number of firms*, not the long-run price:

$$\text{demand } Q_d' = 150 - 2P \;\Rightarrow\; \text{same } P = 30, \text{ more firms } (30 \text{ not } 20).$$

Symmetrically, a demand slump below the level supporting $\min AC$ drives exit until price recovers to 30.

Day 3 checkpoint

You can now

- explain why a competitive firm takes price as given and sets $P = MC$;
- build MC, AVC, AC and read off shutdown (15) and break-even (30) prices;
- derive firm supply $q = (P - 15)/5$ and aggregate it to $Q_s = 4P - 60$;
- connect $PS = \pi + FC$ and sum firm surplus to the market's 450;
- run the long-run logic: profit ($P = 35$, $\pi = 17.5$) $\Rightarrow$ entry $\Rightarrow$ 30 firms at $P = 30$, $\pi = 0$.

Carried into Day 4

drop assumption 1 (many sellers): a single firm faces the *whole* downward-sloping demand, sets $MR < P$, and the efficiency of today's benchmark breaks.

In-class exercises — Day 3

Time for the in-class exercises.

- **Exercise 1** — a firm's cost curves: find MC, AVC, AC, the shutdown and break-even prices, derive short-run supply, and decide operate-or-shut at given prices.
- **Exercise 2** — entry dynamics: a demand boom, short-run profit with a fixed number of firms, then the long-run number of firms at zero profit.

Solutions are at the end of the exercise sheet — try both fully before turning there.